PROMPT FOR RESOLVING THE HADAMARD MATRIX PROBLEM AT ORDER 668

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ABSTRACT. This document contains a full frontier-model research prompt for resolving the first currently unknown Hadamard order. The target is an exact 668×668 sign matrix H satisfying $HH^{\mathsf{T}} = 668I$, or a complete proof that no such matrix exists. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

A Hadamard matrix of order n is a matrix $H \in \{-1, +1\}^{n\times n}$ whose rows are pairwise orthogonal:

$$HH^{\mathsf{T}} = nI_n.$$

Every known necessary condition permits order $668 = 4 \cdot 167$, but no Hadamard matrix of this order is presently known. Resolve the order-668 problem completely:

Either construct $H \in \{-1, +1\}^{668\times 668}$ with $HH^{\mathsf{T}} = 668I$, or prove that none exists.

Assume for purposes of this task that a complete resolution is reachable. Do not assume the positive or negative direction, although the positive direction has the shorter certificate.

Complete-resolution standard

A complete solution must establish exactly one of the following alternatives:

- Positive alternative: provide all 668^2 entries of a sign matrix and prove by exact integer arithmetic that every row has squared norm 668 and every two distinct rows have dot product zero;
- Negative alternative: prove that no order-668 Hadamard matrix exists, with every reduction and computational case split covered by formal argument and independently checkable certificates;
- in either direction, account for all equivalence operations used, including row and column permutations and sign changes, and prove that normalization or symmetry breaking loses no solutions.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a 64-modular Hadamard matrix, a complex Hadamard matrix, a weighing matrix, a conference matrix, or any object whose orthogonality holds only modulo an integer;
- a near-Hadamard matrix, a matrix with small but nonzero correlations, a floating-point candidate, or a matrix of the wrong order;
- a partial block construction whose remaining compatibility conditions are unproved;
- a randomized search log without a complete matrix and exact verifier;
- proofs for infinitely many nearby orders that do not include 668;
- a nonexistence search restricted to one construction family without proving that every Hadamard matrix of order 668 belongs to that family.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- Williamson, propus, Goethals-Seidel, Baumert-Hall, supplementary-difference-set, cocyclic, group-developed, and block-circulant constructions;
- periodic complementary sequences and exact autocorrelation constraints of length 167, exploiting $668 = 4 \cdot 167$;
- lifting or deforming known modular order-668 structures into exact integer orthogonality, while identifying the precise obstruction equations;
- SAT, SMT, CP-SAT, pseudo-Boolean, and mixed-integer formulations with aggressive but proved symmetry reduction;
- local-search, simulated-annealing, evolutionary, reinforcement-learning, and differentiable relaxations used only as generators for exact discrete candidates;
- finite groups of order related to 668, difference sets in group rings, cocycles, orbit matrices, and representation-theoretic decompositions;
- switching operations, trades, bordering, tensor or product constructions, and transformations from designs, codes, or strongly regular graphs;
- for a possible negative route, arithmetic, determinant, Smith-normal-form, p-adic, coding-theoretic, and association-scheme obstructions that apply to all matrices, not just a chosen ansatz.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- confirm the artifact has exactly 668 rows and 668 entries per row, each entry exactly $+1$ or -1;
- compute $HH^{\top}$ in exact integer arithmetic with two independent implementations and compare every one of the 668^2 entries;
- check that file parsing, line endings, delimiters, and sign conventions cannot silently change entries;
- if the matrix is normalized, verify normalization was obtained only by valid equivalence operations

and not by altering orthogonality;

- if constructed from blocks or sequences, verify the fully expanded matrix rather than relying only on derived formulas;
- if nonexistence is claimed, audit exhaustiveness of all equivalence classes and validate each SAT/SMT/ILP certificate with a small independent checker;
- compare the candidate against known equivalence invariants to identify whether it is genuinely new and to aid independent reproduction.

Required final artifact

The final response must include a proof and an archival package. For a positive solution, the matrix itself is the central certificate. Include:

- the complete matrix in plain-text CSV and at least one compact binary or compressed representation;
- a SHA-256 hash of every artifact;
- a short verifier that reads the file, checks dimensions and signs, and computes $HH^{\top}$ exactly;
- a second independently written verifier or a proof-assistant certificate;
- a mathematical derivation explaining the construction, rather than only a raw search output;
- for a negative solution, the complete proof plus all exhaustive-search certificates, generators, checkers, and case manifests.

Public search may be used for standard Hadamard constructions, equivalence theory, and the published status of order 668. Do not search for a purported hidden matrix of order 668 or return merely because the order is listed as open.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.